

Strassen Fast Multiplication Algorithm

$$\text{Let } A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}, B = \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}, \text{ where } A, B \in M(N \times N) \text{ and } A_{ij}, B_{ij} \in M\left(\frac{N}{2} \times \frac{N}{2}\right) \text{ for } i, j \in \{1, 2\}$$

$$\text{Hence, } C = A \times B = \begin{bmatrix} A_{11} \times B_{11} + A_{12} \times B_{21} & A_{11} \times B_{12} + A_{12} \times B_{22} \\ A_{21} \times B_{11} + A_{22} \times B_{21} & A_{21} \times B_{12} + A_{22} \times B_{22} \end{bmatrix} \text{ with time complexity } O(N^3)$$

$$\text{Let } C_1 = (A_{11} + A_{22}) \times (B_{11} + B_{22})$$

$$\text{Let } C_2 = (A_{11} + A_{12}) \times B_{22}, C_3 = A_{11} \times (B_{12} - B_{22})$$

$$\text{Let } C_4 = (A_{21} + A_{22}) \times B_{11}, C_5 = A_{22} \times (B_{21} - B_{11})$$

$$\text{Let } C_6 = (A_{21} - A_{11}) \times (B_{11} + B_{12})$$

$$\text{Let } C_7 = (A_{12} - A_{22}) \times (B_{21} + B_{22})$$

$$\text{Hence, } C = A \times B = \begin{bmatrix} C_1 - C_2 + C_5 + C_7 & C_2 + C_3 \\ C_4 + C_5 & C_1 + C_3 - C_4 + C_6 \end{bmatrix} \text{ with time complexity } O(N^{\log_2 7}) \approx O(N^{2.807})$$